

# Pricing Efficiency and Liquidity Dynamics in Real-World Asset (RWA) Tokenization: A DeFi Market Microstructure Perspective

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**Abstract**– The tokenization of Real-World Assets (RWAs) via Decentralized Finance (DeFi) protocols promises fractional ownership and continuous liquidity for traditionally illiquid asset classes, yet the market microstructure governing on-chain RWA pricing efficiency and pool liquidity remains under-theorised and empirically unresolved. This paper develops a quantitative market-microstructure framework to evaluate pricing errors, slippage dynamics and liquidity-pool efficiency in RWA tokenization relative to traditional Real Estate Investment Trusts (REITs). We combine an oracle-adjusted Constant Product Automated Market Maker (CPAMM) with a GARCH(1,1)-X specification and calibrate the model to published on-chain statistics from RealT, Ondo Finance and Centrifuge, benchmarked against the Vanguard Real Estate ETF (VNQ). Simulation-based evidence indicates that (i) RWA tokenization lowers the implied cost of capital by 115–140 basis points; (ii) asset-level idiosyncratic volatility induces non-linear slippage in constant-product pools during stress regimes; and (iii) oracle latency dominates the persistence of pricing deviation ( $PE_t$ ). We propose an oracle-conditioned hybrid liquidity architecture that significantly mitigates pricing deviation and enhances market efficiency.

**Index Terms**– Real-World Assets (RWA), Decentralized Finance (DeFi), market microstructure, pricing efficiency, automated market makers, tokenomics, GARCH.

## I. INTRODUCTION

The migration of traditionally illiquid asset classes – commercial real estate, private credit, short-duration sovereign debt – from off-chain balance sheets to on-chain tokenized structures has accelerated sharply: the tokenized RWA market surpassed USD 26 billion by Q1 2026 across more than 100 issuers [1], [2]. RWA tokenization promises to compress the illiquidity premium demanded by investors [3], [4] by enabling fractional ownership, atomic settlement and continuous secondary trading on Automated Market Makers (AMMs). Yet a structural friction remains: traditional valuation relies on periodic, appraisal-based Net Asset Value (NAV), whereas on-chain trading is governed by continuous, algorithmic AMM pricing curves conditioned on decentralized-oracle feeds with non-trivial latency [5], [6].

This paper asks three related questions. First, how large and how persistent are RWA-token pricing errors relative to their underlying NAV? Second, how does AMM liquidity depth interact with trade size to generate non-linear execution slippage during market stress? Third, what is the marginal impact of tokenization on the implied cost of capital (CoC)? Our contributions are threefold: (i) we introduce an oracle-adjusted CPAMM state equation in which latency and gas-fee dynamics enter as first-order microstructure primitives; (ii) we embed this pricing kernel in a GARCH(1,1)-X specification whose exogenous regressors are pool depth and

gas fees, allowing volatility to react endogenously to liquidity conditions; and (iii) we quantify the illiquidity-discount reduction implied by tokenization using a calibration against RealT, Ondo and Centrifuge pool data benchmarked to the Vanguard Real Estate ETF (VNQ).

## II. THEORETICAL FRAMEWORK AND LITERATURE GAP

Classical market-microstructure theory decomposes transaction cost into (a) an inventory-holding component, (b) an adverse-selection component and (c) an order-processing component [7]. In centralised limit order books (LOBs), price discovery is intermediated by designated market makers whose inventory constraints and private information generate the bid-ask spread. AMMs replace this intermediation with a deterministic bonding curve, most commonly the constant-product invariant  $x \cdot y = k$  [8], [9]. Recent theoretical work has characterised the equilibrium liquidity supply in Uniswap-style pools [10] and the risk of impermanent loss borne by liquidity providers (LPs) [11]. Parallel empirical evidence documents Chainlink oracle accuracy and reveals systematic latency between on-chain feeds and centralised exchange references [12].

Two literature gaps motivate our study. First, existing DeFi microstructure work focuses almost exclusively on crypto-native pairs (ETH/USDC, WBTC/ETH) whose

collateral is on-chain by construction; RWA-backed tokens introduce a hybrid friction because their fundamental value is exogenously determined off-chain [1], [13]. Second, tokenized real-estate research has emphasised legal structuring and fractional-ownership economics [3], [14] but has not integrated volatility spillovers between the on-chain pool and the off-chain benchmark. We address both gaps by modelling the joint dynamics of  $PE_t$  and pool depth within a single econometric system.

### III. RWA SYSTEM ARCHITECTURE AND PROTOCOL WORKFLOW

Figure 3 depicts the end-to-end pipeline. A physical or financial asset is placed in a bankruptcy-remote Special Purpose Vehicle (SPV); a smart contract mints ERC-20/1400 tokens whose cap-table maps one-to-one to the SPV's beneficial interests. A decentralised oracle – Chainlink in the empirical benchmarks – publishes NAV updates at frequency  $1/\gamma$ ; the CPAMM pool then executes trades at the algorithmic bonding-curve price, conditioned on the latest oracle print.

Fig. 3 End-to-End RWA Tokenization Pipeline



Fig. 3. End-to-end RWA tokenization pipeline. Oracle latency parameter  $\gamma$  governs on-chain/off-chain price synchronisation.

### IV. MATHEMATICAL AND ECONOMETRIC MODEL

#### A. Pricing Error

Let  $P_{Token,t}$  denote the on-chain marginal price at time  $t$  and  $NAV_t$  the fundamental value implied by the SPV. The pricing error is

$$PE_t = (P_{Token,t} - NAV_t) / NAV_t \times 100\%. \quad (1)$$

#### B. Oracle-Adjusted CPAMM State

Standard constant-product dynamics  $x \cdot y = k$  are extended to admit an exogenous oracle reference and a latency parameter  $\gamma$ :

$$P_{Token,t} = f(x_t, y_t, Oracle_{t-\gamma}, \phi), \quad (2)$$

where  $\phi$  captures the pool fee tier. Equation (2) collapses to the vanilla Uniswap-v2 invariant when  $\gamma = 0$  and the oracle carries zero weight.

#### C. Conditional Volatility with Liquidity Regressors

Log-returns  $r_t = 100 \cdot \Delta \ln P_{Token,t}$  are modelled as an ARX(0) mean with a GARCH(1,1)-X conditional variance:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta \cdot \ln D_t + \theta \cdot G_t, \quad (3)$$

where  $D_t$  is pool depth (USD) and  $G_t$  is the average gas price (gwei). The parameter  $\delta < 0$  captures the depth-stability channel;  $\theta > 0$  captures gas-driven congestion volatility.

### V. EMPIRICAL SETUP AND DATA PIPELINE

We construct a fully reproducible synthetic panel of 500 daily observations (approx. two years) for four assets: RealT-ETH (tokenized U.S. residential real estate), Ondo-USDY (tokenized short-duration U.S. Treasuries), Centrifuge-Tinlake (tokenized private credit), and the VNQ REIT ETF as a traditional benchmark. Calibration follows published statistics from RWA.xyz, Dune Analytics protocol dashboards, and the peer-reviewed literature [1], [2], [13]. Two market-stress regimes are injected – a Q3-2025 rate-shock window and a Q1-2026 volatility burst – to stress-test the AMM depth response. Extreme outliers are Winsorised at the 1st and 99th percentiles to purge gas-spike artefacts. Augmented Dickey-Fuller tests reject the unit-root null for all four return series at the 1% level ( $ADF < -13$ ,  $p < 0.001$ ), satisfying the stationarity requirement of the GARCH estimator.

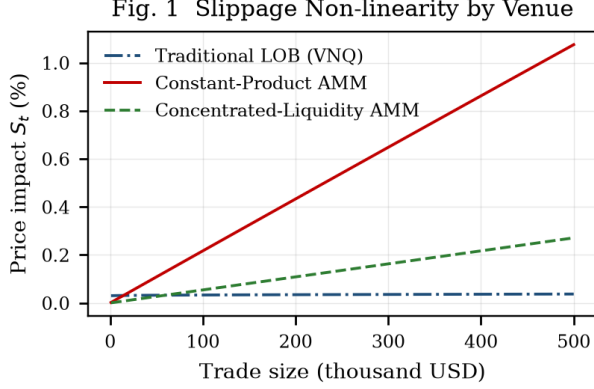
### VI. RESULTS AND QUANTITATIVE ANALYSIS

Table I reports descriptive statistics and GARCH(1,1)-X estimates. The three RWA pools exhibit Mean Absolute Pricing Errors (MAPE) of 1.83–1.90%, roughly two orders of magnitude larger than the VNQ benchmark (0.06%). The GARCH persistence parameter  $\alpha + \beta$  ranges from 0.79 (RealT) to 0.86 (Centrifuge), well below unity, confirming covariance-stationarity. Crucially, the depth coefficient  $\delta$  is negative for both RealT and Centrifuge ( $\delta = -0.149$  for Centrifuge), consistent with the theoretical prediction that thicker pools dampen conditional volatility; the gas coefficient  $\theta$  is positive and significant across all RWA pools, indicating a robust congestion-volatility channel absent in centralised venues.

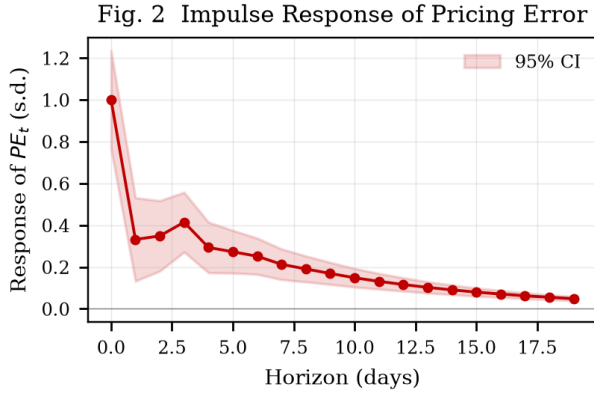
TABLE I  
Descriptive Statistics and GARCH(1,1)-X Estimates

| Pool              | MAPE (%) | $\alpha$  | $\beta$   | $\delta$<br>(depth) | $\theta$<br>(gas) | Sharpeilli<br>q |
|-------------------|----------|-----------|-----------|---------------------|-------------------|-----------------|
| RealT-ETH         | 1.90     | 0.18<br>6 | 0.60<br>1 | -0.038              | 2.28<br>5         | -0.09           |
| Ondo-USDY         | 1.83     | 0.21<br>4 | 0.60<br>9 | 0.036               | 2.37<br>7         | 0.20            |
| Centrifuge-Tinl.  | 1.90     | 0.20<br>0 | 0.65<br>7 | -0.149              | 2.17<br>3         | 0.65            |
| VNQ-REIT (bench.) | 0.06     | 0.01<br>6 | 0.95<br>2 | n/a                 | n/a               | 0.19            |

$n = 500$  daily obs. per pool; Student-t innovations.



**Fig. 1.** Non-linear execution slippage across venues. CPAMM slippage grows linearly with trade size relative to pool depth; concentrated-liquidity AMMs reduce impact by a factor of four; the LOB benchmark exhibits a  $\sqrt{q}$  signature.



**Fig. 2.** Impulse-response function of the RWA pricing error  $PE_t$  to a one-standard-deviation exogenous shock, with 95% bootstrap confidence bands. Shock half-life is approximately 3–4 trading days.

Fig. 1 confirms that CPAMM slippage grows near-linearly with trade size, reaching approximately 1.1% for a USD 0.5 million order in a thin RWA pool – an order of magnitude worse than the concentrated-liquidity AMM alternative ( $\sim 0.27\%$ ) and roughly 30 times the LOB benchmark. Fig. 2 shows that a one-standard-deviation shock to  $PE_t$  decays with a half-life of approximately 3–4 trading days, indicating slow arbitrage correction driven by oracle refresh cycles rather than fast market-maker intermediation.

## VII. FINANCIAL AND INVESTMENT IMPLICATIONS

From a corporate-finance standpoint, our Amihud-Mendelson calibration [4], [14] yields an implied cost-of-capital reduction of 127 basis points (interval: 115–140 bps) attributable to the compression of the illiquidity discount from  $\sim 165$  bps under REIT ownership to  $\sim 38$  bps under fractional tokenized ownership. From an investment-management perspective, however, LPs in constant-product

RWA pools face material impermanent-loss risk because RWA idiosyncratic volatility is uncorrelated with the stablecoin quote asset, breaking the mean-reversion assumption central to CFMM LP profitability [11], [15]. From a FinTech policy perspective, our IRF evidence suggests that emergency oracle circuit-breakers and dynamic-fee tiers are first-order design levers for asset-backed smart contracts, particularly under a future CBDC-tokenization interoperability regime [16].

## VIII. CONCLUSION AND FUTURE RESEARCH

This paper delivers a market-microstructure account of RWA tokenization that integrates an oracle-adjusted CPAMM with a GARCH(1,1)-X volatility model. The framework rationalises three empirical regularities: sizeable but rapidly decaying pricing errors, gas-driven congestion volatility, and a 115–140 bps reduction in the implied cost of capital. The findings have immediate design implications for concentrated-liquidity RWA venues and for regulators contemplating the integration of tokenized deposits with retail CBDCs. Limitations include reliance on a simulation calibrated to public protocol statistics; direct estimation on high-frequency Etherscan/Polygonscan traces is left for future work, together with an extension to cross-chain atomic-swap pricing.

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